AQA Level 3 Certificate — Mathematical Studies (1350)

Correlation & Regression — End of Topic Test 2

Scatter graphs, correlation, regression and prediction

Name: .... Class: .... Date: ....

Time: 70 minutes Total marks: 58

Instructions

- Answer all questions in the spaces provided.
- Use a calculator where appropriate.
- Show all necessary working and give clear interpretations in context; marks for method may be lost if working is not shown.

Question 1

[18 marks]

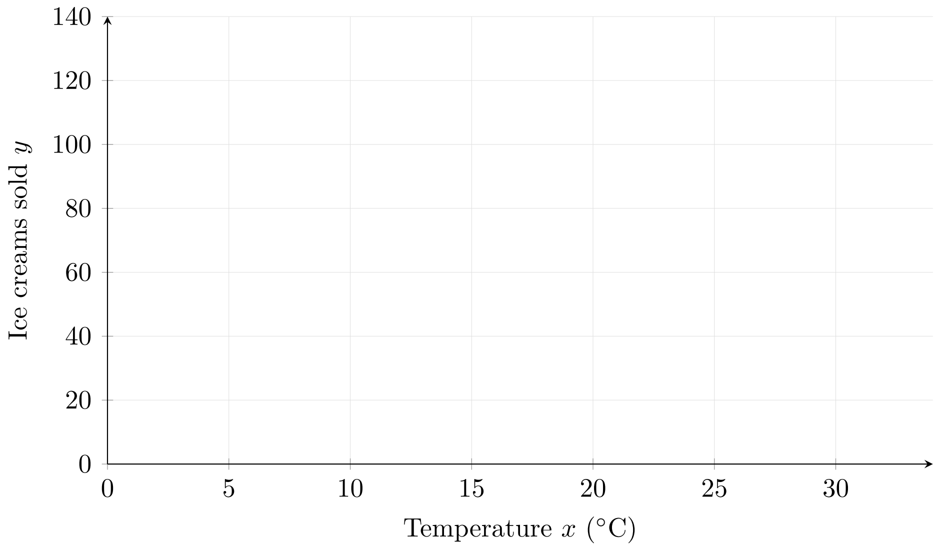
Ice cream sales and temperature

A shop owner records the midday temperature, x (in °C), and the number of ice creams sold, y, on 10 randomly chosen summer days.

Temperature x (°C)	14	18	21	25	28	30	17	22	26	19
Ice creams sold y	40	62	75	95	110	125	55	80	100	65

($n = 10$.)

(a) On the axes provided, plot the 10 points (x, y). Do not draw a line of best fit yet. [3]



(b) Describe the direction and strength of the relationship shown, and give a brief interpretation in context. [3]

(c) Calculate the product-moment correlation coefficient r for these data. Give your answer to 3 decimal places and interpret its value in context. Show key steps or calculator output. [6]

(d) Find the equation of the regression line of y on x in the form $\hat{y} = a + bx$. Use your equation to predict the number of ice creams sold when the temperature is 24°C, stating whether this is interpolation or extrapolation and justifying your answer. [4]

(e) Give two reasons why the regression line might not predict the number of ice creams sold on a particular day perfectly. [2]

Question 2

[14 marks]

Screen time, sleep and stress

A researcher surveys 12 sixth-form students, recording their average daily screen time (in hours), their average nightly sleep duration (in hours), and a self-reported stress score (out of 10, where 10 is most stressed). Two product-moment correlation coefficients are calculated from the data:

- between screen time and sleep duration: $r = -0.82$
- between screen time and stress score: $r = 0.65$

(a) Describe the direction and strength of the relationship between screen time and sleep duration, and interpret it in context. [3]

(b) Describe the direction and strength of the relationship between screen time and stress score, and interpret it in context. [3]

(c) State, with a reason, which of the two relationships is stronger. [2]

(d) The researcher concludes that increasing screen time *causes* higher stress levels in students. Explain why this conclusion is not fully justified by the correlation alone, and suggest one other factor that could help to explain both a student's screen time and their stress score. [4]

(e) Suggest how the researcher's study could be extended or redesigned to provide stronger evidence for a causal link between screen time and stress. [2]

Question 3

[14 marks]

Independent study and GCSE science scores

Two schools each use a linear regression model to predict a student's combined science score, y (out of 100), from the number of hours of independent study per week, x:

$$\text{School P:} \quad \hat{y} = 38 + 3.1x \qquad\qquad \text{School Q:} \quad \hat{y} = 45 + 2.4x$$

(a) Interpret the intercept and the gradient of each model in context, including correct units. [4]

(b) Compare the gradients of the two models. State which school's model predicts a bigger increase in score per extra hour of study, and explain what this means. [3]

(c) A student who studies for 8 hours per week achieves a combined science score of 70. Calculate the residual that would result if this student's result were compared with School P's model, and the residual if compared with School Q's model. Comment on which model fits this result more closely. [4]

(d) Suggest two possible reasons why the two schools' models might differ in this way. [3]

Question 4

[12 marks]

Wingspan and body mass in birds

A researcher measures the wingspan, x (in cm), and body mass, y (in g), of a sample of 9 bird species. A regression line $\hat{y} = -12 + 3.4x$ is fitted, with $r = 0.93$. The wingspans in the sample range from 22 cm to 95 cm. One species in the sample is a particularly large bird whose wingspan and mass are both far greater than the other 8 species.

(a) Use the regression equation to predict the body mass of a bird with a wingspan of 250 cm. Comment on the reliability of this prediction. [3]

(b) Explain how the unusually large species in the sample could have affected the calculated value of r and the fitted regression line. [3]

(c) Give two other limitations of using this simple linear regression model to predict the body mass of bird species in general from their wingspan. [3]

(d) A wildlife blog reports that "a longer wingspan causes birds to be heavier". Explain why the strong correlation ($r = 0.93$) does not prove this claim, suggesting an alternative explanation for the relationship. [3]

END OF TEST